



Relax and Recover

Linux Disaster Recovery best practices with rear



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Who am I

- Independent Unix System Engineer since 1996
- Unix user since 1986
- Linux user since 1991
- Open Source contributor:
 - Make CD-ROM Recovery (mkCDrec)
 - Relax and Recover (rear)
 - SIM Installation and Logging (similar)
 - Adhocracy (adhocr)



Disaster Recovery

- Business Continuity Planning
 - A business continuity plan specifies how a company plans to restore core business operations when disasters occur
- Disaster Recovery
 - Disaster recovery looks specifically at the technical aspects of how a company can get back into operation using backup facilities



Disaster Recovery Concerns

- Uptime
 - Quick restores with minimal or no manual steps after the recovery
- Reliability
 - Avoid corrupted file systems and that system boots after recovery
- Cost
 - DR solutions need to be affordable
- Complexity
 - DR plans tend to be too complex.



Getting started with Relax and Recover (rear)

- Download it from
 - The official tar-balls
 - <https://github.com/rear/rear/downloads/>
 - The rear-snapshot rpm's build from SVN
 - <http://download.opensuse.org/repositories/home:/sschapiro/>
 - The official source
 - <https://github.com/rear/rear>
 - The official repo's (Fedora, EPEL and SLES)
 - yum install rear
 - zypper install rear



Installation of rear

- E.g. on Fedora 17

```
# yum install rear
```

```
Installing:
rear                                noarch                1.13.0-1.fc17                fedora                327 k
Installing for dependencies:
at                                  i686                  3.1.13-7.fc17                fedora                61 k
bc                                  i686                  1.06.95-6.fc17               fedora                106 k
binutils                           i686                  2.22.52.0.1-5.fc17          fedora                3.6 M
ed                                  i686                  1.5-3.fc17                   fedora                72 k
ethtool                            i686                  2:3.2-2.fc17                 fedora                93 k
genisoimage                         i686                  1.1.11-10.fc17              fedora                338 k
...
Install 1 Package (+40 Dependent packages)
Total download size: 21 M
Installed size: 65 M
Is this ok [y/N]: y
```

- **We also need syslinux (and to boot on USB: extlinux)**

```
# yum install syslinux
```
- **Install nfs-utils, cifs-utils, rsync if required**
- **Do not forget openssh(-clients)**



Decide on DR strategy

- Which backup mechanism to use?
 - GNU tar, rsync, bacula, commercial backup program
- Where will the backups reside?
 - NFS share, CIFS share, external USB disk, tape, local spare disk
 - Remote network location
- How shall we start the rescue image
 - Via CDROM (ISO image), tape (OBDR), network (PXE), USB disk

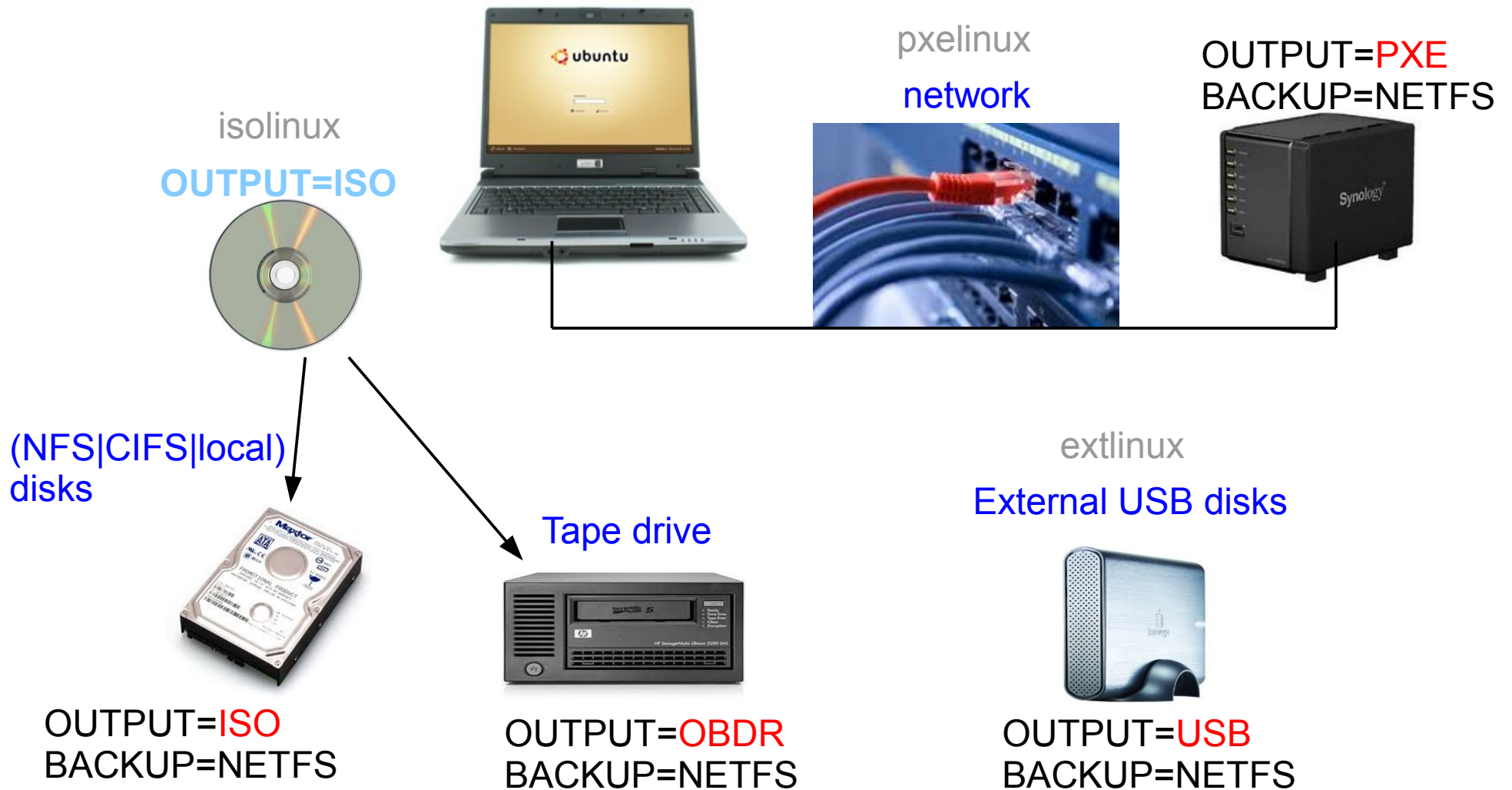


Backup Types

- The major backup types available are
 - **NETFS:** NFS, CIFS, USB, TAPE
 - RSYNC: rsync method
 - REQUESTRESTORE, EXTERNAL
 - BACULA (open source backup software)
 - DP, NBU, TSM, GALAXY[7] (commercial stuff)
- Some not (yet) implemented backup types (waiting on sponsors)
 - NSR (Legato Networker)
 - CDROM



BACKUP type NETFS





Location BACKUP_URL

- **BACKUP=NETFS**
- **BACKUP_URL** can be
 - File type: `BACKUP_URL=file:///directory/`
 - NFS type: `BACKUP_URL=nfs://nfs-server/directory/`
 - CIFS type: `BACKUP_URL=cifs://samba/directory/`
 - USB type: `BACKUP_URL=usb:///dev/sdc1/directory/`
 - Tape type: `BACKUP_URL=tape:///dev/nst0`



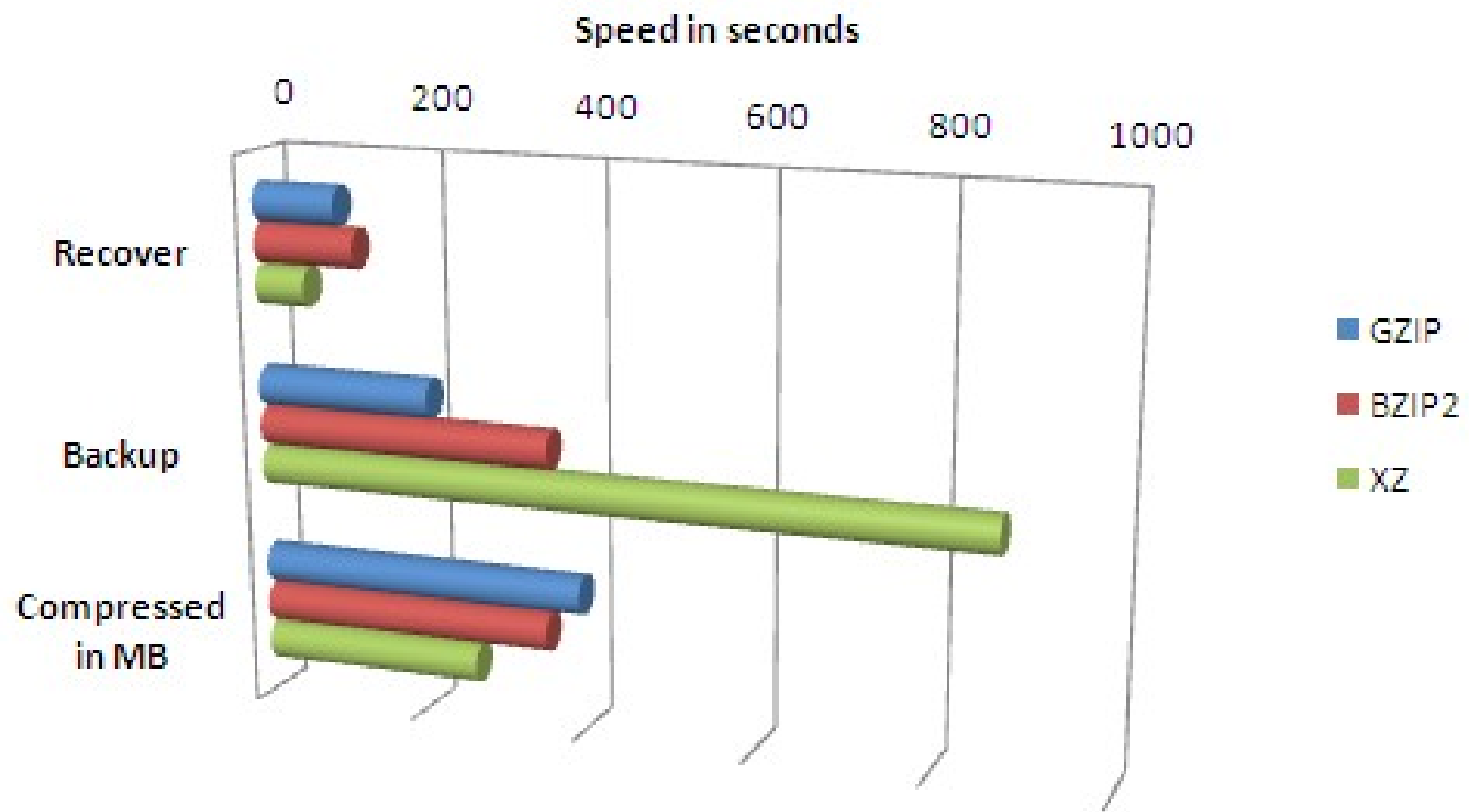
Backup Program

- `BACKUP=NETFS`
- `/usr/share/rear/conf/default.conf`
 - By default is `BACKUP_PROG=tar`
 - However, `BACKUP_PROG=rsync` is possible for local attached storage
 - `BACKUP_PROG_COMPRESS_OPTIONS="--gzip"`
 - `BACKUP_PROG_COMPRESS_SUFFIX=".gz"`
 - `BACKUP_PROG_EXCLUDE=( '/tmp/*' '/dev/shm/*' )`



BACKUP_PROG_COMPRESS_OPTIONS

Rear Backup/Recover tests (NETFS)





/etc/rear/local.conf

- Define your settings in /etc/rear/local.conf (or /etc/rear/site.conf)
- ```
grep -v -E '(^#|^$)' /etc/rear/local.conf
```

```
OUTPUT=ISO
```

```
MODULES_LOAD=(vmxnet)
```
- Add:

```
BACKUP=NETFS
```

```
BACKUP_URL=nfs://server/path
```
- On NFS server backup => /path/**\$(hostname)**/



- View system configuration:

```
rear dump
```

```
Relax and Recover 1.13.0 / $Date$
```

```
Dumping out configuration and system information
```

```
This is a 'Linux-x86_64' system, compatible with 'Linux-i386'.
```

```
System definition:
```

```
ARCH = Linux-i386
```

```
OS = GNU/Linux
```

```
OS_MASTER_VENDOR =
```

```
OS_MASTER_VERSION =
```

```
OS_MASTER_VENDOR_ARCH =
```

```
OS_MASTER_VENDOR_VERSION =
```

```
OS_MASTER_VENDOR_VERSION_ARCH =
```

```
OS_VENDOR = Fedora
```

```
OS_VERSION = 16
```

```
OS_VENDOR_ARCH = Fedora/i386
```

```
OS_VENDOR_VERSION = Fedora/16
```



- Usage: rear [-dDsSvV] [-r KERNEL] COMMAND [-- ARGS...]
- Available options:
  - **-d** debug mode; log debug messages
  - **-D** debugscript mode; log every function call
  - **-r KERNEL** kernel version to use; current: '2.6.42.3-2.fc15.i686.PAE'
  - **-s** simulation mode; show what scripts rear would include
  - **-S** step-by-step mode; acknowledge each script individually
  - **-v** verbose mode; show more output
  - **-V** version information



- Usage: `rear [-dDsSvV] [-r KERNEL] COMMAND [-- ARGS...]`
- List of commands:
  - `checklayout`      check if the disk layout has changed
  - `format`            format and label media for use with rear
  - `mkbackup`          create rescue media and backup system
  - `mkbackuponly`      backup system without creating rescue media
  - `mkrescue`          create rescue media only
  - `recover`            recover the system; only valid during rescue
  - `savelayout`        save the disk layout of the system
  - `shell`              start a bash within rear; development tool





## Disaster Recovery in Practice

- Gather system information
- Store the disk layout
  - Partitioning, LVM and RAID configuration
  - File systems, file system labels ...
  - Boot loader (GRUB, LILO, ELILO)
- Make a system backup (OS and user data)
- Create boot-able rescue media with system configuration (and optional with backup data)
- All steps are done “**online**”



## Rear mkrescue

- Will create an ISO image stored as
  - `/tmp/rear-$(hostname).iso`
  - On NFS server as `/path/$(hostname)/rear-$(hostname).iso`
- Inspect file `/var/lib/rear/layout/disklayout.conf`
- Try to boot from the ISO image into the RESCUE system
  - Use 'dmesg' to check if devices were found



## Rear mkbackup

- Create rescue image with backup archive
- Do not forget to browse through the `/tmp/rear-$ (hostname).log` file for errors





## Recovery Process in detail

- Boot system from rescue media
- Restore disk layout
  - Create partitions, RAID configuration and LVM
  - Create file systems (mkfs, mkswap)
  - Configure file systems (labels, mount points)
- Restore the backup data
- Restore the boot loader
- Inspect & Reboot



## Recover with rear (1)

- Boot rescue image and select 'recover'

```
Relax and Recover v1.13.0

Recover fedora

Other actions
Help for Relax and Recover
Boot First Local disk (hd0)
Boot Second Local disk (hd1)
Boot Next device
Hardware Detection Tool
ReBoot system
Power off system

Press [Tab] to edit, [F2] for help, [F1] for version info

Rear rescue image kernel 3.1.7-1.fc16.i686.PAE Thu, 03 May 2012 14:46:
BACKUP=NETFS OUTPUT=ISO BACKUP_URL=nfs://192.168.1.100:2049/test
```



## Recover with rear (2)

- Wait until you see the login prompt

```
Attempting to start the DHCP client daemon
Running 60-network-devices.sh...
Running 62-routing.sh...
* * * Rescue System is ready * * *

Relax and Recover 1.13.0 / $Date$

Relax and Recover comes with ABSOLUTELY NO WARRANTY; for details see
the GNU General Public License at: http://www.gnu.org/licenses/gpl.html

Host fedora using Backup NETFS and Output ISO
Build date: Thu, 03 May 2012 14:45:18 +0200

Fedora release 16 (Verne)
Kernel 3.1.7-1.fc16.i686.PAE on an i686 (tty1)

fedora login: root

Welcome to Relax and Recover. Run "rear recover" to restore your system

RESCUE fedora:~ # _
```



## Recover with rear (3)

```
RESCUE fedora:~ # rear -v recover
Relax and Recover 1.13.0 / $Date$
Calculating backup archive size
Backup archive size is 460M (compressed)
Comparing disks.
Disk configuration is identical, proceeding with restore.
Start system layout restoration.
Creating partitions for disk /dev/sda (gpt)
Creating LVM PV /dev/sda3
 0 logical volume(s) in volume group "vg_fedora" now active
Restoring LVM VG vg_fedora
Creating ext4-filesystem / on /dev/mapper/vg_fedora-lv_root
Mounting filesystem /
Creating ext4-filesystem /boot on /dev/sda2
Mounting filesystem /boot
Creating swap on /dev/mapper/vg_fedora-lv_swap
Disk layout created.
Restoring from 'nfs://hpx189.ncsbe.eu.jnj.com/test/fedora/backup.tar.gz'
Restored 2078 MiB [avg 7440 KiB/sec] OK
Restored 2078 MiB in 287 seconds [avg 7414 KiB/sec]
Installing GRUB2 boot loader
Installation finished. No error reported.

Finished recovering your system. You can explore it under '/mnt/local'.

Finished in 313 seconds
RESCUE fedora:~ # _
```



## Recover with rear (4)

```
Metadata Sequence No 4
VG Access read/write
VG Status resizable
MAX LV 0
Cur LV 2
Open LV 1
Max PV 0
Cur PV 1
Act PV 1
VG Size 4.50 GiB
PE Size 32.00 MiB
Total PE 144
Alloc PE / Size 144 / 4.50 GiB
Free PE / Size 0 / 0
VG UUID H7VT2i-mvUY-Y2e5-5adL-bzCw-28CE-gb3Y1x

RESCUE fedora:~ # df
Filesystem 1K-blocks Used Available Use% Mounted on
devtmpfs 435664 0 435664 0% /dev
tmpfs 456244 0 456244 0% /dev/shm
tmpfs 456244 200 456044 1% /run
tmpfs 456244 0 456244 0% /sys/fs/cgroup
/dev/mapper/vg_fedora-lv_root 3128548 2473820 527592 83% /mnt/local
/dev/sda2 495844 72805 397439 16% /mnt/local/boot
RESCUE fedora:~ #
```

- Ready? **Reboot (shutdown -r 0)**
- That's it – wait a while for the selinux relabeling
- Verify the restored system





## Cloning with rear (1)

- Start the recover process: `rear -v recover`

```
RESCUE beefy:~ # rear -v recover
Relax and Recover 1.13.0 / $Date$
Comparing disks.
Device sda has size 5368709120, 6442450944 expected
Switching to manual disk layout configuration.
Original disk /dev/sda does not exist in the target system. Please choose an appropriate replacement.
1) /dev/sda ←
2) /dev/sdb
3) Do not map disk.
#? 1
```

```
2012-05-15 12:55:03 Disk /dev/sda chosen as replacement for /dev/sda.
Disk /dev/sda chosen as replacement for /dev/sda.
This is the disk mapping table:
 /dev/sda /dev/sda
Please confirm that /var/lib/rear/layout/disklayout.conf is as you expect.
1) View disk layout (disklayout.conf) 4) Go to Rear shell
2) Edit disk layout (disklayout.conf) 5) Continue recovery
3) View original disk space usage 6) Abort Rear
#? 1
```



## Cloning with rear (2)

```
disk /dev/sda 6442450944 gpt
part /dev/sda 1048576 1048576 rear-noname bios_grub /dev/sda1
part /dev/sda 524288000 2097152 ext4 boot /dev/sda2
part /dev/sda 4843372544 526385152 rear-noname lvm /dev/sda3
disk /dev/sdb 4294967296 gpt
lvmdev /dev/vg /dev/sda3 WIV8Xr-hN1o-JNRn-XMUU-K16I-I0tF-ErxYUv 11552768
lvmgrp /dev/vg 32768 176 5767168
lvmvol /dev/vg lv_swap 50 3276800
lvmvol /dev/vg lv_root 126 8257536
fs /dev/mapper/vg-lv_root / ext4 uuid=53faa99e-be97-4a15-80d9-936a0103e33e label=
= blocksize=4096 reserved_blocks=4% max_mounts=-1 check_interval=0d options=rw,r
elative,seclabel,user_xattr,barrier=1,data=ordered
fs /dev/sda2 /boot ext4 uuid=576e6373-50c9-4762-8bbd-95f83931a680 label= blocksi
ze=1024 reserved_blocks=5% max_mounts=-1 check_interval=0d options=rw,relative,s
eclabel,user_xattr,barrier=1,data=ordered
swap /dev/mapper/vg-lv_swap uuid=bf30769d-f25b-4dfd-bd2a-cecf4694e02a label=
/var/lib/rear/layout/disklayout.conf (END)
```

```
1) View disk layout (disklayout.conf)
2) Edit disk layout (disklayout.conf)
3) View original disk space usage
4) Go to Rear shell
5) Continue recovery
6) Abort Rear
#? 5
```

Partition rear-noname on /dev/sda: size reduced to fit on disk.  
Please confirm that `/var/lib/rear/layout/diskrestore.sh` is as you expect.

```
1) View restore script (diskrestore.sh)
2) Edit restore script (diskrestore.sh)
3) View original disk space usage
4) Go to Rear shell
5) Continue recovery
6) Abort Rear
#? 1
```



## Cloning with rear (3)

```
#!/bin/bash

LogPrint "Start system layout restoration."

mkdir -p /mnt/local
if create_component "vgchange" "rear" ; then
 lvchange -a n >&8
 component_created "vgchange" "rear"
fi

set -e
set -x

if create_component "/dev/sda" "disk" ; then
Create /dev/sda (disk)
Log "Erasing MBR of disk /dev/sda"
dd if=/dev/zero of=/dev/sda bs=512 count=1
sync
LogPrint "Creating partitions for disk /dev/sda (gpt)"
parted -s /dev/sda mklabel gpt >&2
parted -s /dev/sda mkpart rear-noame 32768B 1081343B >&2
parted -s /dev/sda set 1 bios_grub on >&2
parted -s /dev/sda mkpart ext4 1085440B 525373439B >&2
parted -s /dev/sda set 2 boot on >&2
/var/lib/rear/layout/diskrestore.sh_

1) View restore script (diskrestore.sh)
2) Edit restore script (diskrestore.sh)
3) View original disk space usage
4) Go to Rear shell
5) Continue recovery ←
6) Abort Rear
#? 5
```



## Cloning with rear (4)

```
Start system layout restoration.
Creating partitions for disk /dev/sda (gpt)
Creating LVM PV /dev/sda3
 0 logical volume(s) in volume group "vg" now active
Creating LVM VG vg
Creating LVM volume vg/lv_swap
Creating LVM volume vg/lv_root
An error occurred during layout recreation.

1) View Rear log ←
2) View original disk space usage
3) Go to Rear shell
4) Edit restore script (diskrestore.sh)
5) Continue restore script
6) Abort Rear
#? 1
```

```
+++ echo -e 'Creating LVM volume vg/lv_root'
+++ lvm lvcreate -l 126 -n lv_root vg
Volume group "vg" has insufficient free space (94 extents): 126 required.
2012-05-15 13:08:28 An error occurred during layout recreation.
```

```
1) View Rear log
2) View original disk space usage
3) Go to Rear shell
4) Edit restore script (diskrestore.sh) ←
5) Continue restore script
6) Abort Rear
#? 4
```

```
LogPrint "Creating LVM volume vg/lv_root"
lvm lvcreate -l 126 -n lv_root vg >&2
component created "/dev/mapper/vg-lv_root"
```

```
lvm lvcreate -l 94 -n lv_root vg >&2
```



## Cloning with rear (5)

```
1) View Rear log
2) View original disk space usage
3) Go to Rear shell
4) Edit restore script (diskrestore.sh)
5) Continue restore script ←
6) Abort Rear
#? 5_
```

```
#? 5
Start system layout restoration.
Skipping /dev/sda (disk) as it has already been created.
Skipping /dev/sda1 (part) as it has already been created.
Skipping /dev/sda2 (part) as it has already been created.
Skipping /dev/sda3 (part) as it has already been created.
Skipping pv:/dev/sda3 (lvmdev) as it has already been created.
Skipping /dev/vg (lvmgrp) as it has already been created.
Skipping /dev/mapper/vg-lv_swap (lvmvol) as it has already been created.
Creating LVM volume vg/lv_root
Creating ext4-filesystem / on /dev/mapper/vg-lv_root
Mounting filesystem /
Creating ext4-filesystem /boot on /dev/sda2
Mounting filesystem /boot
Creating swap on /dev/mapper/vg-lv_swap
Disk layout created.
Please start the restore process on your backup host.

Make sure that you restore the data into '/mnt/local' instead of '/' because the
hard disks of the recovered system are mounted there.

Please restore your backup in the provided shell and, when finished, type exit
in the shell to continue recovery.
rear>
```



## Cloning with rear (6)

```
RESCUE beefy:~ # cat /etc/rear/local.conf
sample local configuration

Create Rear rescue media as ISO image
OUTPUT=ISO

optionally define (non-default) backup software, e.g. TSM, NBU, DP, BACULA
BACKUP=TSM

the following is required on older VMware VMs
MODULES_LOAD=(vmxnet)

to see boot messages on the serial console (uncomment next line)
KERNEL_CMDLINE="console=tty0 console=ttyS1"
RESCUE beefy:~ # df
Filesystem 1K-blocks Used Available Use% Mounted on
devtmpfs 425332 0 425332 0% /dev
tmpfs 453128 0 453128 0% /dev/shm
tmpfs 453128 204 452924 1% /run
tmpfs 453128 0 453128 0% /sys/fs/cgroup
/dev/mapper/vg-lv_root 3072888 111288 2838396 4% /mnt/local
/dev/sda2 508745 23411 459734 5% /mnt/local/boot
RESCUE beefy:~ #
```

So you better know what you're doing, right?  
The **BACKUP** variable was not set in the `/etc/rear/local.conf` configuration file!



Get your hands dirty?

- We hope you want to dig deeper into rear!
- Getting started:
  - Use: `rear -s mkbackup`  
to see the flow of the scripts it will execute
  - Depends on BACKUP method, architecture and OS version/brand
  - Be careful: `rear -s recover`  
follows a different flow (seems logically, but you must understand the difference)



Where is the code?

- Main script is `/usr/sbin/rear`
- All the other scripts live under `/usr/share/rear`
- Documentation is at `/usr/share/doc/rear-X.y.Z`
- ***Good news! It's all written in Bash***





## Where to put a script?

- **mkbackup** method: `/usr/share/rear/...`
  - `conf/` - configuration files (`/etc/rear/*.conf` read last)
  - `prep/` - preparation work; checking the environment
  - `layout/save/` - save the disk layout `/var/lib/rear/layout`
  - `rescue/` - modules, network, storage,...
  - `build/` - populate the initial ramdisk for our rescue image
  - `pack/` - create the initrd and copy kernel
  - `output/` - create the ISO image and copy to `OUTPUT_URL`
  - `backup/` - make the backup archive to `BACKUP_URL`



## Where to put a script? (2)

- **recover** method: `/usr/share/rear/...`
  - `conf/` - read the configuration file + `/etc/rear/*.conf`
  - `setup/` - user defined scripts to run before recover
  - `verify/` - to check if a recover is possible at all
  - `layout/prepare` – recreate the disk layout
  - `restore/` - restore the archive from `BACKUP_URL`
  - `finalize/` - do some dirty tricks for disks, grub,...
  - `wrapup/` - copy the recover log to `/mnt/local/root/`



## Example script: sysreqs.sh

- A simple script to save basic system requirements – sysreqs.sh
  - OS version; rear version
  - CPU, memory
  - Disk space requirements
  - IP addresses in use; routes
- Copy sysreqs.sh to a flow, e.g. rescue is a good choice
  - `# cp /tmp/sysreqs.sh \`  
`/usr/share/rear/rescue/GNU/Linux/96_sysreqs.sh`



## Example script: sysreqs.sh

- A simple script to save basic system requirements – sysreqs.sh
  - OS version; rear version
  - CPU, memory
  - Disk space requirements
  - IP addresses in use; routes
- Copy sysreqs.sh to a flow, e.g. rescue is a good choice
  - `# cp /tmp/sysreqs.sh \`  
`/usr/share/rear/rescue/GNU/Linux/96_sysreqs.sh`



## Test the script

- `# rear -s mkrescue | grep sysreqs`  
Source rescue/GNU/Linux/96\_sysreqs.sh
- `# rear -v mkrescue`
- `# cat /var/lib/rear/sysreqs/Minimal_System_Requirements.txt`

```
fedora - 2012-05-22 11:26
```

```
Operating system:
LSB Version: :core-4.0-ia32:core-4.0-noarch
Distributor ID: Fedora
Description: Fedora release 16 (Verne)
Release: 16
Codename: Verne
```

```
Relax and recover version:
Relax and Recover 1.13.0 / $Date$
```

```
There are 1 CPU core(s) at 2393.832 MHz
748 MiB of physical memory
```

```
Disk space requirements:
OS (vg + swap + /boot)
size: 6.09 GiB
Additional VGs
```

```
Network Information:
IP addresses:
ip 6 ::1 subnet /128 scope host DNS name
ip 192.168.5.135 subnet /24 DNS name
ip 6 fe80::20c:29ff:fe63:5cd0 subnet /64 scope link DNS name
```



## Test the script

- `# rear -s mkrescue | grep sysreqs`  
Source rescue/GNU/Linux/96\_sysreqs.sh
- `# rear -v mkrescue`
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```
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LSB Version: :core-4.0-ia32:core-4.0-noarch
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Release: 16
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```

```
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There are 1 CPU core(s) at 2393.832 MHz
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```

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Network Information:
IP addresses:
ip 6 ::1 subnet /128 scope host DNS name
ip 192.168.5.135 subnet /24 DNS name
ip 6 fe80::20c:29ff:fe63:5cd0 subnet /64 scope link DNS name
```



<https://github.com/rear/rear/issues>

Code

Network

Pull Requests 0

Issues 48

Wiki

Graphs

Browse Issues

Milestones

Search: Issues & Milestones... 

New Issue

Everyone's Issues 48

Assigned to you 6

Created by you 1

Mentioning you 1

No milestone selected 

Labels

SF 7

bug 10

cleanup 12

collaboration 3

discuss 24

documentation 1

feature 19

feedback 4

Manage Labels

New label name

No active filters. Use the sidebar to filter issues. Keyboard shortcuts available 

48 open issues

32 closed issues

Submitted

Updated

Comments

Close

Label

Assignee

Milestone

1 2 Next »

  #80 **Add "StopIfFail" function to \_input-output-functions.sh**  
by djembe65 17 hours ago

  #79 **Items to discuss at LinuxTag 2012 meeting on 24/5 at 10am** collaboration discuss  
by dagwieers 2 days ago 3 comments

  #78 **epel missing dependencies**  
by lesmikesell 4 days ago 1 comment

  #76 **CIFS parameters not correctly passed**  
by r0bby 15 days ago 1 comment

  #75 **DP: press any key** feature SF  
by jhoekx a month ago

  #73 **Problem in lib/compatibility-functions.sh because blkid shows no output when there are no partitions** bug  
by kpieth a month ago

  #72 **Manpage still states sourceforge as project homepage** bug  
by baccenfutter a month ago

  #71 **swap partitions should not be resized** bug  
by kpieth a month ago

  #69 **Improve, redesign or remove "rear validate"** cleanup discuss  
by dagwieers a month ago



## Contacts

**Web-site:** <http://rear.github.com/>

**GitHib:** <https://github.com/rear/rear>

**Mailing list:** [rear-users@lists.sourceforge.net](mailto:rear-users@lists.sourceforge.net)

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